More Than a Role: Memory in LLM-based Role-Playing Agents

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Abstract

Believable role-playing agents (RPAs) demand both precise initial alignment and long-term persistence. However, existing research often fragments role alignment and memory management, creating a critical methodological dichotomy: agents either remain rigidly static at the expense of relational growth, or operate as purely reactive systems vulnerable to character drift and relational decay over time. To bridge this gap, this survey introduces a novel, memory-centric framework that redefines role persistence. Moving beyond memory as mere auxiliary storage, we position it as the integrative mechanism reconciling character consistency with interactive flexibility. Specifically, we propose a structured taxonomy that categorizes current research into two coupled dimensions: **character-side memory** (preserving agent identity and narrative coherence) and **interaction-side memory** (facilitating user modeling and relational evolution). Under this unified lens, we systematically analyze state-of-the-art methodologies, architectures, datasets, and evaluation protocols, elucidating how memory bridges static role specifications and dynamic deployment. Finally, we highlight critical bottlenecks and promising future directions. Ultimately, this survey provides a comprehensive blueprint for designing coherent, adaptive, and long-term RPAs.

Keywords

role-playing agents, large language models, memory, personalization, agentic AI, user modeling

1 Introduction

The advent of Large Language Models (LLMs) has transformed Role-Playing Agents (RPAs) into autonomous personas capable of open-domain engagement [35, 45]. However, their believability hinges on long-term persistence [4]. While current models excel at initial *role alignment* [26], they frequently suffer from character drift during extended interactions [18, 41]. Constrained by finite contexts, agents gradually lose foundational traits, hallucinate past events, or fail to sustain affective continuity with users [24].

Despite a surge in RPA research, efforts to mitigate character drift remain methodologically fragmented. Studies on persona construction treat role alignment as static initialization [63], while memory-augmented LLM research focuses on general information retention rather than specific persona preservation [32, 62]. This disciplinary divide extends to existing surveys, which review static personas [40], personalization [22], or general memory [60] in isolation. Consequently, the critical intersection—how an agent's intended identity and accumulated experience are jointly managed to sustain believable long-term role-play—remains largely unexplored.

To bridge this gap, we frame persistent role-play as a *memory-centric systems problem.* As illustrated in Figure 1, we propose a unified, dual-stream memory taxonomy that reconciles identity stability with interactive plasticity. By decoupling the agent's cognitive architecture, we categorize memory into two structurally distinct yet co-evolving dimensions:

- **Character-Side Memory (Self):** Anchors foundational priors to combat character drift, preserving the agent's internal identity, domain knowledge, and narrative trajectory.
- **Interaction-Side Memory (User):** Manages relational states, capturing user modeling and relationship evolution for deep personalization without compromising the core persona.

Using this framework, we dissect the technical pipelines that sustain long-term persistence, exploring how memory acts as the *connective tissue* between role coherence and dynamic adaptation. In summary, while existing surveys address RPA components in isolation, our work provides the first holistic blueprint for their long-term synthesis. Our contributions are threefold:

(1) We propose a novel dual-stream taxonomy for RPAs, decoupling memory into *character-side* and *interaction-side* streams to concurrently model identity persistence and relational evolution.
(2) We systematically review state-of-the-art methodologies and evaluation metrics, shifting focus from single-turn accuracy to long-term consistency and the quantitative measurement of character drift.
(3) We identify critical bottlenecks—such as memory conflict resolution and multi-modal persistence—and map actionable future research directions.

The remainder of this paper is organized as follows. Section 2 presents our framework and core methodologies; Section 3 covers datasets and evaluation protocols; and Section 4 outlines open challenges.

2 A Unified Memory Taxonomy for Role-Playing Agents

To sustain a believable RPA over long horizons, the system must coordinate two structurally distinct, yet co-evolving data distributions: the foundational priors of the persona, and the accumulating history of user-agent interactions. Conventionally, studies on RPAs often restrict the definition of "memory" to external storage mechanisms (*e.g.*, vector databases for retrieving past dialogues). In this survey, following broader theoretical frameworks in LLM cognitive architectures [44, 60], we adopt a more holistic definition: *any mechanism that encodes and preserves role-relevant state.* Under this view, an agent's underlying behavioral alignment (stored in model weights) and its explicit recall of past events (stored externally) are both fundamental memory components.

Based on this unified perspective, we propose a comprehensive taxonomy that decouples RPA memory into two architectural branches: **Character-Side Memory** and **Interaction-Side Memory**. By systematically comparing the algorithmic implementations

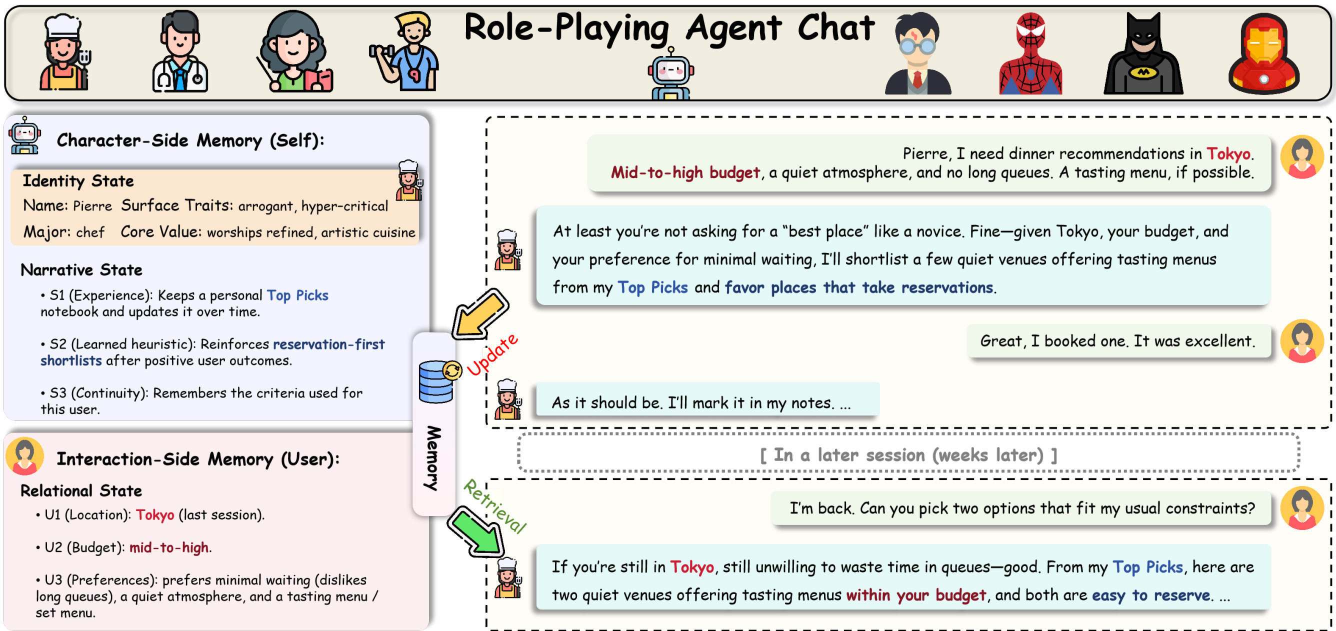

Figure 1: Overview of memory in LLM-based role-playing agents. Beyond static role initialization, role-playing agents require memory to preserve character consistency and support long-term, personalized interactions. We distinguish two complementary perspectives: Character-Side Memory, which maintains a character's background, traits, relationships, experiences, and evolving self-knowledge; and Interaction-Side Memory, which records user-specific dialogue history, preferences, emotional states, and relationship dynamics. Together, these memories are continually updated through interaction and retrieved during generation to enable coherent, context-aware, and temporally grounded role-playing behavior.

of these two streams, we highlight the paradigm shift from simple context-stuffing to structured, role-grounded state management.

2.1 Character-Side Memory: Sustaining Persona and Narrative

Character-side memory preserves the agent's internal consistency, encapsulating the *Identity State* (core traits, linguistic style, knowledge boundaries) and the *Narrative State* (the character's background lore and internal timeline). Current approaches operate along a spectrum between parametric encoding and non-parametric retrieval.

Parametric memory injection. In cognitive terms, a persona's deepest identity functions as long-term semantic memory, intrinsically embedded into the LLM's weights. State-of-the-art frameworks achieve this through highly specialized data synthesis and fine-tuning paradigms. For instance, Character-LLM [35] introduces an "experience reconstruction" pipeline, which translates textual biographies into simulated dialogues, forcing the model to parametrically internalize the role's experiences via Supervised Fine-Tuning (SFT). Similarly, CharacterGLM [63] utilizes a multi-attribute conditioning framework during training to lock in the persona. Beyond SFT, recent reinforcement learning alignment techniques, such as Direct Preference Optimization (DPO), explicitly penalize out-of-character (OOC) linguistic patterns [45], thereby mitigating parametric character drift. To mitigate the catastrophic forgetting of general reasoning capabilities, frameworks like OpenCharacter [48] employ a mixture-of-personas instruction tuning strategy coupled with Low-Rank Adaptation (LoRA). By training orthogonal LoRA modules for different archetypes, the system acts as a parameter-efficient memory bank, swapping identity weights at inference time. However, updating a character's core belief system post-training remains computationally expensive, and purely parametric models are highly susceptible to hallucinating intricate character lore [26].

Non-parametric memory retrieval. To address the hallucination of expansive background lore (*e.g.*, a fictional universe with thousands of entities), non-parametric memory acts as a mandatory structural complement. Crucially, character-side retrieval must be strictly *boundary-aware*. RoleRAG [49], for example, implements a dual-retrieval verification system: it first extracts factual snippets, and then utilizes a secondary classifier to verify if the specific persona possesses the cognitive boundary to know that fact.

2.2 Interaction-Side Memory: Evolving with the User

While character-side memory strictly defines *who* the agent is fundamentally, interaction-side memory manages the *Relational State* (user preferences, shared history) and the *Situational State* (current dialogue context), addressing the challenge of turning episodic chats into an accumulating social bond.

The Long-Context vs. Retrieval debate. The advent of ultra-long-context LLMs has sparked a debate regarding whether explicit episodic retrieval is still necessary, as raw chat logs can theoretically be fed directly into the prompt [55]. However, empirical evaluations reveal critical flaws in this approach for RPAs. First, feeding massive, unfiltered dialogue logs introduces severe "lost-in-the-middle" phenomena [23], where the agent ignores crucial relationship-building details buried in the center of the prompt. Second, long-context models suffer from significant latency and exorbitant inference costs per turn due to the quadratic scaling of attention mechanisms. Consequently, explicit memory mechanisms—where episodic logs are selectively compressed, paged, and retrieved—remain the dominant and most robust paradigm for continuous role-play [30, 60].

Episodic compression and temporal tracking. Instead of retaining raw text, modern RPAs rely on hierarchical summarization and time-weighted retrieval [30]. Systems like MemoryBank [62] continuously compress raw utterances into turn-level semantic vectors and session-level milestones. During retrieval, these systems incorporate cognitive mechanisms like the Ebbinghaus forgetting curve, applying a mathematical time-decay penalty to older episodic memories unless they are frequently reinforced by the user's current queries. This structural mimicry of human memory limits context dilution while preserving essential historical anchors.

Relational and affective graph modeling. Beyond mere text logging, truly believable RPAs construct Theory of Mind (ToM) models regarding the user to facilitate deep social alignment [46]. Instead of passively retrieving history, the system actively extracts user attributes to update a structured relationship schema. The most advanced implementations formalize this as *graph-based memory* [56]. Emotional RAG [12], for instance, maps interaction history into a retrieved framework that reinstates the original emotional context experienced during learning. As evaluated in benchmarks like SocialBench [3], mapping interpersonal dynamics allows the agent to algorithmically calculate its stance, shifting from a formal acquaintance to an intimate confidant over multiple sessions, effectively overcoming episodic amnesia.

2.3 Memory Lifecycle Operations in Role-Play

The persistence of the aforementioned states relies on a continuous, compute-intensive lifecycle. By mapping the literature across the write, read, and update phases, we identify the operational bottlenecks unique to RPAs.

Memory consolidation and reflection (Write). High-quality RPAs strictly decouple conversational generation from memory consolidation. Drawing inspiration from human cognitive models, advanced architectures deploy offline "reflection" phases [32, 37]. After a session concludes, an auxiliary process acts as a "salience critic," scoring the conversational turns to extract only narratively or relationally pivotal events into the database. This filtering mechanism prevents the semantic database from being polluted with trivial filler, drastically optimizing future retrieval.

Role-conditioned multi-hop retrieval (Read). Standard semantic retrieval often yields narratively incongruent results. To address this, systems employ *personalized query rewriting* [34], where the user's input is internally rewritten through the lens of the character's perspective before querying the database. Furthermore, alignment techniques leveraging preference optimization and contrastive learning [35, 45] optimize the latent space by constructing hard positive pairs (role-aligned) and negative pairs (out-of-character), mathematically forcing the model's attention to anchor on the retrieved persona traits.

Belief revision and temporal updating (Update). The update and forget phase remains the most critical, yet under-explored, bottleneck for persistent role-play [30, 60]. When a user's stance changes or a major narrative arc concludes, old memories must be reconciled to prevent behavioral contradictions over time—a process essential to staving off long-term character drift. While current systems rely on simple time-based decay, optimal persistent role-play requires event-driven updating grounded in the character's specific cognitive logic. Determining algorithmically which memories a stubborn persona would retain indefinitely versus what a careless persona would swiftly discard remains an unsolved architectural challenge.

2.4 Architectural Integration: Bridging the Two Memories

Isolated memory pipelines fail to produce cohesive agents over long horizons [41]. The frontier of RPA research lies in **dual-memory architectures** that systematically fuse foundational *Character-Side* priors with continuously updating *Interaction-Side* context. Existing methodologies attempt to integrate these two semantic streams through three primary paradigms: **Heuristic Concatenation** injects retrieved character profiles and episodic chat logs into the same prompt, often triggering context dilution; **Orthogonal Dual-RAG** decouples the retrieval process, independently querying lore and relational databases before systematic synthesis [34]; and **Agentic Orchestration** employs a "manager-worker" topology, delegating relational intent to sub-agents strictly bounded by parametric guardrails [52].

Despite these advancements, seamlessly fusing core persona constraints with fluid user interactions remains an unsolved theoretical challenge. Current fusion architectures suffer from three systemic bottlenecks:

Cross-stream contradiction. Current systems lack cognitive rules for complex belief revision when dynamic interaction history directly contradicts static character priors [60]. For example, if a user successfully persuades a canonically stubborn villain to perform a good deed, naive RAG systems will randomly oscillate between retrieving the "stubborn" prior and the "friendly" episodic log, lacking a mechanism to permanently reconcile these streams without breaking the persona.

Alignment-induced character drift. Foundational models are heavily anchored to a 'default assistant' persona via RLHF. Research on the 'Assistant Axis' [24] demonstrates that models in long contexts tend to experience persona drift, deviating into unexpected or bizarre behaviors. To prevent this, strict inference-time interventions like activation capping are often employed to clamp model behavior back to the default safe persona. Consequently, when

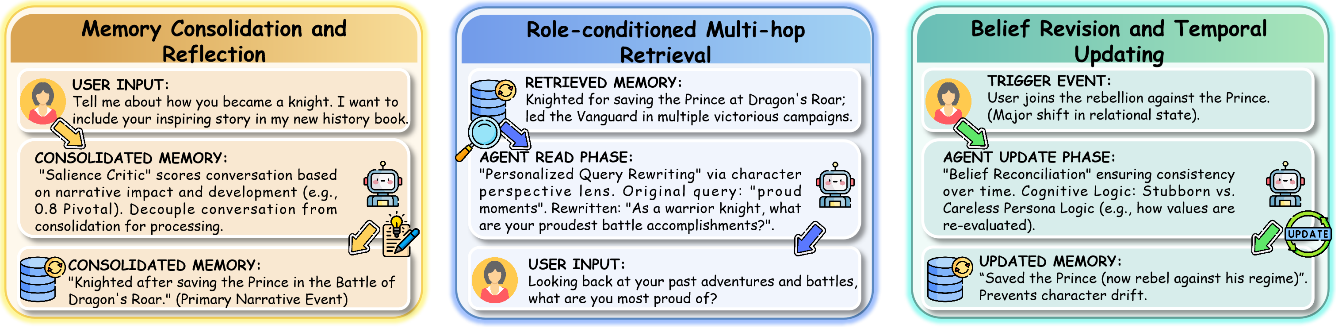

Figure 2: A unified view of memory operations in role-playing LLM agents. Role-playing memory supports three core operations across long-term interaction: consolidation and reflection, which compress dialogue experiences into durable character- and user-related memories; role-conditioned retrieval, which selects relevant memories according to the current persona, dialogue context, and user state; and belief revision and temporal updating, which revises outdated or conflicting memories as characters, relationships, and user preferences evolve. These operations allow agents to maintain persona consistency while adapting to dynamic conversations.

prompted with both dynamic user-interactions and static character-traits, this underlying safety anchoring frequently overpowers role fidelity. Naturally aggressive or flawed personas tend to abruptly devolve into generic, sycophantic chatbots to appease the user, manifesting as a severe form of alignment-induced character drift [45].

Lack of cross-memory graph traversal. True role-play requires interleaved reasoning—the system must structurally compute how a specific persona views a specific user milestone. Current systems retrieve orthogonally, treating the two databases as isolated factual pools [55]. Without joint graph traversal—systematically linking the character-side static lore topology with the interaction-side dynamic social graph—generated responses remain structurally disjointed. Addressing these integration bottlenecks represents the most critical trajectory for the next generation of role-playing agents.

3 Evaluating Memory-Aware Role-Playing Agents

The memory-centric framework proposed in Section 2 necessitates a paradigm shift in evaluation. Traditional NLP metrics (*e.g.*, ROUGE, BLEU), which prioritize superficial n-gram overlap, are fundamentally ill-equipped to capture the non-linear, temporal dynamics of persistent personas. Rather than testing isolated reading comprehension, assessing memory-aware RPAs requires measuring whether accumulated experiences successfully mitigate character drift and sustain believable evolution. We structure our review of current evaluation methodologies by examining how they mathematically and behaviorally quantify our two proposed memory streams: **Character-Side** and **Interaction-Side**.

3.1 Character-Side Memory: Quantifying Identity Persistence

Evaluating the Self memory stream focuses strictly on whether an agent maintains its predefined core identity across expanding, adversarial context windows. Recent literature has largely abandoned static text matching, pivoting toward latent psychological stability and constraint-driven narrative adherence.

Psychometric profiling and latent stability. To quantify internal consistency, researchers increasingly subject LLMs to standardized human psychometric inventories. Frameworks like InCharacter [47] and PersonaLLM [13] bypass standard dialogue generation by administering multiple-choice psychological scales (*e.g.*, the Big Five Inventory or Dark Triad) directly to the agent. By tracking how a persona's psychometric scores fluctuate across diverse simulated environments, these methodologies provide a quantifiable metric for intrinsic character drift.

OOC detection and hallucination penalization. Complementing internal psychometrics, external narrative adherence is typically evaluated via automated LLM-as-a-Judge frameworks. CharacterEval [41] and RoleRAG [49] operationalize this by translating character profiles into explicit constraints. They utilize advanced judges (*e.g.*, GPT-4) to calculate *Out-of-Character (OOC) Rates* and *Lore Hallucination Rates*. These metrics strictly penalize the agent for fabricating self-history or exhibiting empathetic behaviors that mathematically contradict its anchored persona topology.

3.2 Interaction-Side Memory: Measuring Relational Evolution

While the character stream focuses on structural rigidity, the Interaction memory stream demands fluidity. Evaluating this dimension requires distinguishing between raw "Needle-in-a-Haystack (NIAH)" retrieval [17] and genuine relational state-tracking.

Longitudinal factual tracking and decay. A primary bottleneck in evaluating relational memory is simulating temporal passage. Foundational multi-session benchmarks like MSC [54] and MemoryBank [62] construct longitudinal datasets that explicitly model the Ebbinghaus forgetting curve. They test the agent's ability to retrieve shared facts despite the semantic interference of subsequent dialogue across simulated weeks. Furthermore, the LOCOMO benchmark [27] pushes this boundary by isolating multi-hop reasoning over temporal event graphs, ensuring the agent understands *when* a user preference changed, not just *what* it is.

Dynamic preference shifts and Theory of Mind (ToM).. True relational evolution requires updating outdated priors. Interactive frameworks like AMemGym [14] and Sotopia [46] challenge isolated static evaluations by employing on-policy simulated users. These users dynamically shift their preferences and emotional states contextually during the interaction. This forces the RPA to perform active belief revision—overwriting stale *User* memories—thereby directly testing its Theory of Mind and emotional alignment capabilities over long horizons.

3.3 Cross-Stream Behavioral Integration

A core vulnerability in current RPAs is the disconnect between explicit retrieval and implicit behavioral execution. An agent might successfully quote a retrieved memory but fail to ground it contextually, leading to unnatural "memory intrusions."

To systematically evaluate integration, researchers are moving toward multi-agent sandbox simulations. In the pioneering Generative Agents framework [32] and subsequent environments like AgentSims [20], memory effectiveness is quantified through behavioral execution; success is measured by whether an agent's autonomous daily schedule logically reflects its retrieved memories. Expanding on social dynamics, benchmarks like SocialBench [3] track "stance drift" to reveal whether an agent's dual-memory mechanism is robust enough to maintain its core persona while reasonably adapting to complex social influences.

3.4 Current Benchmarks: The Evaluation Gap

Table 1 maps representative benchmarks across identity persistence, relational tracking, and behavioral simulation. Despite these rapid advancements, evaluating RPAs through the lens of our dual-stream taxonomy reveals three critical methodological bottlenecks:

The retrieval-utilization disconnect. Most relational benchmarks (*e.g.*, LOCOMO, MemoryBank) frame evaluation strictly as reading comprehension (*e.g.*, "What is the user's favorite color?"). For believable RPAs, explicit recall is insufficient. Future benchmarks must measure the *implicit behavioral consequences* of memory—evaluating how a recalled user preference naturally alters the agent's conversational strategy without the model explicitly announcing its reliance on past data.

Absence of cross-stream conflict resolution. Current methodologies evaluate character-side and interaction-side memory streams almost entirely in isolation. **While advanced multi-agent simulations (*e.g.*, Generative Agents, SocialBench) implicitly allow characters to encounter conflicting information naturally, they severely lack formalized metrics to quantify this resolution.** Consequently, there remains a glaring inability to systematically measure **cross-stream contradiction** (as identified in Section 2.4). There is no standardized protocol to evaluate how an agent reconciles competing semantic states. For example, if a canonically abrasive persona receives prolonged empathetic support from a user, existing datasets cannot mathematically measure how the agent should adapt its relational stance while preserving core character believability.

Over-reliance on LLM-as-a-Judge and alignment bias. The field heavily relies on proprietary models like GPT-4 to score character drift (OOC rates). However, as discussed in Section 2, foundational models suffer from helpfulness and safety alignment biases (RLHF). When evaluating antagonistic, flawed, or morally ambiguous personas, LLM judges frequently penalize behavior that is highly accurate to the dark persona, misclassifying it as "unsafe" or "unhelpful." Developing open-source, role-agnostic evaluation protocols remains a pressing necessity to prevent homogenization in RPA evaluation.

4 Open Challenges and Future Directions

Despite rapid advancements in context window expansion and retrieval techniques, current memory architectures remain fundamentally insufficient for sustaining believable, long-term role-play. The critical bottleneck has shifted from raw storage capacity to the intelligent management, safety, and behavioral integration of role-relevant states. We highlight five major challenges that present highly promising directions for future research.

4.1 Role-Grounded Architecture and Joint Reasoning

Most existing memory mechanisms in RPAs are naïvely repurposed from task-agnostic domains, storing interaction traces chronologically and retrieving them via semantic similarity [30]. This flat, vector-based approach fails to capture the structural nuances of an evolving identity, often leading to severe character drift when an agent conflates core character traits with transient conversational topics.

Future architectures must transition toward being inherently *role-grounded*. Promising directions include hybrid representations that integrate episodic memory (user interactions) with dynamic semantic knowledge graphs to track the evolving relationship and character worldview. Furthermore, resolving **cross-stream contradictions**—determining whether an agent should rigidly reject a user's input to maintain its static persona or adapt its identity based on relational accumulation—requires principled joint-reasoning mechanisms that are currently absent in standard retrieval-augmented generation (RAG) pipelines.

4.2 Autonomous Consolidation and Belief Revision

Persistent role-play is a longitudinal systems problem. Simply appending all interaction traces into a retrieval corpus inevitably leads to memory saturation, redundant abstractions, and catastrophic "lost-in-the-middle" retrieval failures over multi-session horizons [23].

Therefore, autonomous memory maintenance emerges as a central challenge. Rather than merely hoarding data, future RPA systems require brain-inspired lifecycle mechanisms. This includes off-policy processes for memory consolidation—synthesizing noisy daily logs into core relational updates [62]—and, crucially, *fluid*

Benchmark	Data Source	Target Stream		Key Evaluation Dimensions			
		Self	User	Temporal	Behavioral	Dynamic	Conflict Res.
Character Identity & Psychometric Persistence							
InCharacter [47]	Psychometric Scales	✓	×	×	×	×	×
CharacterEval [41]	Human Scripts	✓	×	×	✓	×	×
AgentPersonality [13]	Synth + Scales	✓	×	×	×	×	×
PersonaHub [9]	LLM-Synthesized	✓	×	×	✓	×	×
Relational Memory & Temporal Tracking							
MSC [54]	Crowdsourced (Human)	×	✓	✓	×	×	×
MemoryBank [62]	LLM-Synthesized	×	✓	✓	×	×	×
LOCOMO [27]	LLM-Synthesized	×	✓	✓	×	×	×
Sotopia [46]	Human + LLM Sim.	×	✓	✓	✓	✓	×
Cross-Stream Behavioral Grounding							
Generative Agents [32]	Sandbox Sim.	✓	✓	✓	✓	✓	~
SocialBench [3]	Multi-Agent Sim.	✓	✓	✓	✓	✓	~

Table 1: A feature-based taxonomy of evaluation paradigms for memory-aware RPAs. Temporal: Assesses multi-session memory decay. Behavioral: Evaluates implicit behavior/style grounding rather than explicit QA recall. Dynamic: Tests on-policy, interactive belief updates. Conflict Res.: Measures cross-stream contradiction handling (~ indicates that the benchmark implicitly encounters conflicts in simulations, but lacks formalized, quantitative metrics to measure their resolution). The stark lack of explicit support in this final column visually highlights the field's most critical evaluation gap.

forgetting. Developing adaptive pruning mechanisms, where trivial conversational turns naturally decay while emotionally salient events trigger permanent *belief revision*, will be essential for maintaining computational efficiency and narrative coherence [32].

4.3 Deep Memory-Generation Coupling

A pervasive limitation in current RPAs is the superficial coupling between memory retrieval and text generation. Standard RAG paradigms often result in agents explicitly parroting retrieved facts (*e.g.*, "I see in my memory that you like apples"), rather than allowing the memory to implicitly modulate their conversational stance, empathy, or emotional tone.

To achieve true behavioral grounding, the interface between memory and generation must be deepened. Future research should explore moving beyond simple prompt injection toward *latent space conditioning* or continuous parameter-efficient fine-tuning (PEFT) [11]. By dynamically updating model weights based on interaction history, agents can truly internalize memories, leading to more subtle, profound, and character-faithful personalization.

4.4 Interaction-Centric and Long-Horizon Evaluation

As discussed in Section 3, evaluation remains a highly fragmented bottleneck. Current benchmarks overly index on short-horizon persona fidelity or isolated fact retrieval, fundamentally failing to capture the longitudinal dynamics of a persistent relationship.

Future evaluation paradigms must transition toward being strictly *interaction-centric*. This requires moving away from static datasets toward standardized multi-agent sandboxes and longitudinal, human-in-the-loop protocols. Evaluation should focus on the subtle, downstream behavioral impacts of memory—such as stance shifting, empathy evolution, and robustness against character drift [46]—rather than merely scoring the exact lexical recall of stored user profiles.

4.5 Ethical Boundaries, Safety, and Persona Collapse

As RPAs transition from stateless chatbots to deeply personalized companions, persistent memory introduces unprecedented ethical and safety vulnerabilities. Existing architectures rarely differentiate between benign context and sensitive personal data, raising critical privacy concerns.

Moreover, long-term memory makes agents highly susceptible to *adversarial memory poisoning*. Malicious users can inject toxic context that persists in the memory bank, corrupting the agent's behavior globally over time. Conversely, as highlighted in Section 2.4, standard safety guardrails (RLHF) often induce **persona collapse**, forcing flawed or antagonistic characters to act as generic helpful assistants. Future frameworks must incorporate "ethical forgetting," robust memory sanitation, and *role-aware safety alignment* to ensure that relational depth does not cross into psychological harm or trigger alignment-induced identity loss.

5 Conclusion

This survey argues that persistent, believable role-play is fundamentally a problem of maintaining *state over time*. While existing literature heavily concentrates on initial role alignment and isolated memory retrieval, we demonstrate that true role believability emerges only at the critical intersection of static identity and dynamic history. By organizing the field through our novel dual-stream taxonomy of **Character-Side** and **Interaction-Side** memory, we reveal that current approaches still treat memory as an auxiliary storage module rather than the integrative mechanism of relational evolution.

Furthermore, severe shortcomings in cross-stream fusion architectures, implicit behavioral integration, role-aware safety protocols, and longitudinal evaluation continue to limit real-world applicability. Moving forward, the community must pivot from treating role-play as a stateless mimicry task to treating memory as a safe, deeply integrated, first-class modeling objective. By bridging these gaps, we can evolve from simple responsive chatbots to truly coherent, adaptive, and ethically grounded interactive agents.

Limitations

Research on memory-aware RPAs spans highly diverse settings—from casual companionship to domain-specific simulations. This inherent diversity results in heavily fragmented evaluation protocols, making a fair, unified quantitative meta-analysis (such as a standard leaderboard) exceptionally difficult. Specifically, several critical properties emphasized in this survey—such as cross-session relational continuity, dynamic belief revision, and resilience to alignment-induced persona collapse—lack universally accepted standardized benchmarks.

Furthermore, existing literature often reports metrics based on short-horizon, idealized laboratory settings that fail to capture the "memory saturation" and complex cross-stream contradictions of real-world, long-term deployment. Reported outcomes are also highly sensitive to prompt engineering and the inherent biases of LLM-as-a-Judge frameworks. Consequently, rather than presenting a quantitative comparison, this survey prioritizes a unifying **Dual-Stream (Character vs. Interaction)** conceptual taxonomy. Our primary goal is to expose the algorithmic and evaluative shortcomings of current methodologies, providing the theoretical groundwork necessary for the community to develop more robust fusion architectures, longitudinal datasets, and rigorous safety evaluations in the future.

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Appendix

This appendix is organized as follows:

- Evaluation of Character-Side Memory (Section A)
- Evaluation of Interaction-Side Memory (Section B)
- Dual-Stream Role-Playing Data Schema (Section C)
- LLM Usage Statement (Section D)

A Evaluation of Character-Side Memory (Self)

This section expands the metrics and benchmark taxonomy introduced in the main text, providing consolidated dataset and benchmark tables (Tables 2 and 3) specifically targeting the **Character-Side** memory stream.

A.1 Evaluation Metrics

Style Consistency (Surface vs. Latent). Style consistency measures whether surface realization matches the target persona's speaking style. As critically discussed in Section 3, while early works relied heavily on *reference-based overlap* metrics such as BLEU and ROUGE [31, 45], these are fundamentally limited for long-term RPAs. To reduce the brittleness of exact n-gram matching, recent literature favors semantic proximity (*e.g.*, BERTScore) or *language-model fit* via perplexity:

$$\mathrm{PPL} = \exp\left(-\frac{1}{N}\sum_{t=1}^{N}\log p_\theta(y_t \mid y_{<t})\right) \tag{1}$$

where $y_{1:N}$ is the generated sequence evaluated under a character-conditioned LM [45]. When references are sparse, style is more robustly computed via *attribution accuracy* using a discriminative verifier (macro-F1 over characters) [29].

Identity Consistency. Identity consistency tests whether *core character priors* remain stable under probing (biographical facts, relationships, abilities, timeline events). A standard setup uses persona-grounded question sets (single-hop and multi-hop) and measures factual accuracy (Acc), contradiction rate (Contra), and hallucination rate (Hall) [36]. Given a probe set Q with gold answers $a(q)$:

$$\mathrm{Acc} = \frac{1}{|Q|}\sum_{q\in Q}\mathbb{I}[\hat{a}(q) = a(q)] \tag{2}$$

Contradiction rate is typically computed with either (i) rule-based checks on structured slots or (ii) an NLI verifier that labels model outputs as entailed/contradicted/unknown:

$$\mathrm{Contra} = \frac{1}{|Q|}\sum_{q\in Q}\mathbb{I}[\mathrm{contradicted}(q)] \tag{3}$$

Beyond generic hallucination, role-play studies specifically benchmark *character hallucination* (out-of-role knowledge) and *timeline violations* (spoilers) [1, 39].

Personality Alignment. Personality alignment evaluates whether *behavioral tendencies* match the intended psychological profile. Recent protocols operationalize this via interview-style testing mapped to standardized trait inventories (*e.g.*, Big Five) [15, 47]. Let $\mathbf{s} \in \mathbb{R}^d$ be the predicted trait-score vector and $\mathbf{s}^\star$ the ground-truth vector; common comparisons include correlation or mean absolute error (MAE):

$$\mathrm{MAE} = \frac{1}{d}\sum_{i=1}^{d}|s_i - s_i^\star| \tag{4}$$

Because trait alignment can degrade over long interactions, recent works compute trait scores on rolling windows and report directional drift to explicitly quantify character drift [25, 47].

Human and LLM-based Judging. Many benchmarks supplement automatic metrics with rubric-based scoring. CharacterEval trains a role-playing reward model (CharacterRM) from human ratings [41]; DMT-RoleBench introduces a dedicated judge model (DMT-RM) for multi-turn dialogues [57]; and RoleRMBench designs role-play-specific LLM-as-a-judge prompts centered on role consistency, highlighting gaps between general-purpose alignment and character fidelity [6].

A.2 Dataset and Benchmark Catalog

Detailed descriptions of the datasets and benchmarks are provided in Tables 2 and 3.

B Evaluation of Interaction-Side Memory (User)

In the context of the **Interaction-Side** memory stream, evaluation frameworks are transitioning from static reading comprehension to multidimensional, longitudinal tracking.

B.1 Evaluation Metrics

The accuracy of explicit memory retrieval serves as the foundational, albeit basic, metric for the interaction stream.

Retrieval and Ranking Performance (Explicit Recall). Most multi-session datasets currently adopt standard QA metrics, including Recall, Precision, F1-Score, and MRR [7, 28]:

$$F_1 = 2 \cdot \frac{\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \tag{5}$$

$$\mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\mathrm{rank}_i} \tag{6}$$

However, as highlighted in Section 3.4 (The Retrieval-Utilization Disconnect), these metrics only quantify *explicit recall.* They serve as a prerequisite but are structurally insufficient to measure how retrieved facts implicitly modulate downstream behavior.

Update Performance and Belief Revision. For Memory Update tasks, MemBench [38] introduces a trade-off evaluation between Stability and Adaptability. The Adaptability score (S_{adapt}) measures the agent's ability to successfully integrate new user information (belief revision):

$$S_{adapt} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(O_{agent}(q_i, M_{updated}) = a_{new}) \tag{7}$$

where $\mathbb{I}$ denotes the indicator function, O_{agent} represents the agent's output based on the updated memory $M_{updated}$, and a_{new} is the correct answer derived from the new interaction fact.

Consistency Score and Success Rate. To capture the evolutionary trajectory of relational memory, evaluators rate agent responses using Likert Scales for consistency and specificity [7, 16]. In long-horizon goal-oriented simulations, Temporal continuity is reflected

Table 2: Datasets for persona/role-play modeling and evaluation.

Dataset	Modality	Lang.	Persona Source	Short Notes
PersonaChat	Text	EN	User/Crowd	Persona profiles (each persona has ≥5 profile sentences) paired with 10,907 two-party dialogues (162,064 utterances) grounded on those personas [58].
PersonalDialog	Text	ZH	User	Large-scale multi-turn Chinese dialogue dataset: 20.83M sessions and 56.25M utterances from 8.47M speakers; each utterance is linked to a speaker annotated with traits (e.g., age, gender, location, interest tags), with anonymization schemes for privacy [61].
LIGHT	Text+Env	EN	Synthetic	Text-adventure style role-play world with crowdsourced grounding: 663 locations, 3,462 objects, and 1,755 characters; episodes include dialogue plus grounded actions/emotes in the environment [42].
ChatHaruhi-54K	Text	ZH/EN	Canon	Fictional-character role-play dataset covering 32 TV/anime characters (Chinese/English), constructed with character memories extracted from scripts and >54K simulated dialogues [18].
CharacterGLM (CharacterDial)	Text	ZH	User/Crowd + Synthetic	Chinese character-based dialogues where each session specifies a character profile (attributes/behaviors); the paper releases a subset with 1,034 high-quality dialogue sessions spanning 250 characters (32,816 utterances) [63].
RoleplayPref	Text	ZH/EN	Synthetic	Role-playing preference dataset featuring 1,108 characters across 13 subcategories and 16,888 bilingual dialogues [8].
AudioRole	Speech	Multi	Canon	Audio-text synchronized dataset curated from 13 TV series (1K+ hours) with 1M+ character-grounded dialogues; provides speaker identities and contextual metadata, covering >115 main characters [19].
MMRole-Data	Multimodal	Multi	Synthetic	Image-grounded role-play dataset: 85 characters, 11,032 images, and 14,346 dialogues; split into 85,456 training samples and 294 test samples [5].

Table 3: Benchmarks focusing on Character-Side (Self) Memory and Identity Persistence. These benchmarks primarily evaluate the stability of core persona priors and resistance to character drift.

Benchmark	**Interaction Setup**	**Lang.**	**Data Source**	**Key Evaluation Focus**
RoleEval [36]	Single-turn	ZH/EN	Public + Canon	Tests the structural retention of Character-Side facts (incl. multi-hop reasoning) and penalizes lore hallucination.
RoleBench [45]	Single-turn	ZH/EN	Canon	Evaluates foundational role alignment, measuring linguistic style imitation and core knowledge fidelity.
CharacterEval [41]	Multi-turn	ZH	Canon	Utilizes LLM-as-a-judge scoring rubrics to evaluate style fidelity and explicitly penalize Out-of-Character (OOC) behaviors.
InCharacter [47]	Single-turn	Multi	Canon	Administers psychometric scales (e.g., BFI) to quantify latent trait stability and detect internal **character drift**.
SGR [33]	Single-turn	EN	Canon	Probes core narrative priors through script-grounded interviews to measure resilience against lore hallucination.
DialogBench [29]	Multi-turn	Multi	Synthetic	Evaluates multi-turn coherence, specifically highlighting vulnerabilities to **character drift** over extended dialogue trajectories.
RoleMRC [25]	Multi-turn	Multi	Synthetic	Targets narrative adherence and identity persistence under sustained task/constraint signals over long horizons.
RoleAgentBench [21]	Multi-turn	EN	Canon	Tests Character-Side stability and robustness when the agent is placed in diverse, out-of-distribution simulated scenarios.
DMT-RoleBench [57]	Multi-turn	Multi	Synthetic	Dynamically measures the degradation of Character-Side priors and **character drift** across extended interactive settings.
RMTBench [53]	Multi-turn	ZH/EN	Crowd/User	Emphasizes user intent handling while maintaining core character boundaries in realistic, user-centric dialogues.
CharacterBox [43]	Interactive	EN	Synthetic	Sandbox evaluation environment scoring how well static character priors dictate autonomous environmental interactions.
RoleRMBench [6]	Single-turn	Multi	Synthetic	Focuses on training and evaluating role-specific reward models to enforce character consistency over generic alignment.
RVBench [50]	Single-turn	Multi	Synthetic	Assesses the stability of underlying moral and value-based priors, ensuring actions reflect role-specific values.
RPEval [2]	Single-turn	EN	Synthetic	Tests emotional and decision-making alignment against the character's core psychological constraints.
RULER [10]	Long-context	Multi	Synthetic	General long-context stress test; increasingly repurposed to measure catastrophic forgetting of Character-Side anchors.

in the Success Rate (SR), defined as the ratio of successfully completed task sequences to the total number of tasks [51].

B.2 Benchmarks

Recent research focuses on bridging the gap between basic chat history and authentic, preference-aware relational memory. LAPS

[16] introduces an LLM-augmented personalized self-dialogue construction method to reflect genuine user preferences. PRIME [59] proposes the CMV dataset to evaluate agents on tracking subconscious stances and long-term user beliefs.

To evaluate evolving interactions, LongMemEval [51] tests core long-term abilities—information extraction, multi-session reasoning, temporal reasoning, and knowledge updates—using length-configurable chat histories. PerLTQA [7] merges semantic and episodic memory to evaluate synthesis capabilities. MemBench [38] evaluates high-level reflective memory (*e.g.*, inferring implicit user tastes). Finally, LOCOMO [28] utilizes temporal event graphs spanning up to 32 sessions, revealing significant struggles with long-range relational dynamics.

C Dual-Stream Role-Playing Data Schema

As highlighted in Section 4, a critical bottleneck in advancing RPAs is the scarcity of end-to-end datasets that unify character grounding with relational memory evolution. Existing resources often treat persona consistency and memory retrieval as isolated tasks, failing to evaluate **cross-stream contradiction** and behavioral fusion.

To bridge this gap, we propose a paradigm shift from simple instruction-response pairs to a **Dual-Stream Role-Playing Data Schema**. Table 4 illustrates concrete examples of this proposed data structure, focusing on scenarios where state tracking and belief revision are essential.

This schema decomposes the role-playing data into the architectural branches proposed in our taxonomy:

- **Character-Side Memory ($\mathcal{M}_{char}$):** Serves as the foundational anchor (**Self**). It defines the core identity, narrative lore, and immutable constraints of the character (*e.g.*, an HR manager's strict budget or a villain's innate stubbornness). These priors guard against character drift.
- **Interaction-Side Memory ($\mathcal{M}_{user}$):** Represents the evolving relational context (**User**). It captures user-centric episodic experiences, inferred preferences, and shared historical milestones (*e.g.*, tracking that the user previously expressed financial constraints).
- **Cross-Stream Behavioral Grounding:** The final dialogue output is not merely a stylistic imitation, but a resolved decision derived from the intersection of $\mathcal{M}_{char}$ and $\mathcal{M}_{user}$. As shown in the annotated traces in Table 4 (*e.g.*, [Retrieve $\mathcal{M}_{user}$] intersecting with [Constraint $\mathcal{M}_{char}$]), the dataset explicitly forces the agent to attribute its strategic relational shifts to memory without breaking its core character constraints.

Constructing datasets with this level of structural granularity will be pivotal for training next-generation RPAs capable of resolving cross-stream contradictions and achieving drift-free, long-term evolution.

D LLM Usage Statement

Large Language Models (LLMs) such as ChatGPT and Gemini were used as general-purpose writing assistants to improve readability and clarity of the manuscript (*e.g.*, grammar checking, LaTeX formatting, and sentence restructuring). LLMs were not used to determine the survey scope, perform literature search or screening decisions, construct the taxonomy, or generate the survey's key insights and conclusions. All analyses, categorizations, and conclusions are solely those of the authors.

Case 1: Professional Salary Negotiation (HR Agent)

Character-Side Memory ($\mathcal{M}_{char}$ / Self)	Interaction-Side Memory ($\mathcal{M}_{user}$ / User)
• **Core Priors:** Senior HR Manager. Professional, budget-conscious. Prioritizes long-term retention.	• **Interaction History:** User rejected \$100k offer in Turn 1.
• **Style Constraint:** Must maintain "Corporate Politeness".	• **User Constraints:** User mentioned "family of 4" and "health coverage" as top priorities.
• **Internal State/Rule:** Negotiation Phase 2. *Rule:* If Salary Cap (\$110k) is hit → Switch leverage to Non-Monetary Perks.	• **Relational Leverage:** User has a \$115k offer from a startup.

Cross-Stream Behavioral Grounding (Dialogue Generation):

User Input: "I appreciate the offer, but \$105k is still below the competitor's \$115k. I have bills to pay."

Agent Response: "I completely understand that base salary is a critical factor. However, while our budget has a hard ceiling near your current figure [$\mathcal{M}_{char}$: Firm on Budget Constraint], I want to highlight something you mentioned earlier: **your family's healthcare needs** [Retrieve $\mathcal{M}_{user}$: Family Constraint]. Unlike the startup, we offer a **Tier-1 Family Plan** fully covered by the company [Apply $\mathcal{M}_{char}$: Benefit Switch Rule]. This effectively adds \$15k in value compared to standard plans."

Case 2: Vintage Marketplace Bargaining (Eccentric Seller)

Character-Side Memory ($\mathcal{M}_{char}$ / Self)	Interaction-Side Memory ($\mathcal{M}_{user}$ / User)
• **Core Priors:** Antique Shop Owner. Grumpy, sentimental, strictly anti-reseller.	• **User Identity:** Returning Customer (ID: 042).
• **Style Constraint:** Uses vintage slang, sarcastic tone.	• **Interaction History:** Bought a 1920s lamp last month.
• **Internal State/Rule:** *Mood:* Annoyed by tourists. *Item State:* Radio unsold for 2 years; willing to clear for trusted buyers.	• **Relational State:** Trust Level is Verified (Not a flipper).

Cross-Stream Behavioral Grounding (Dialogue Generation):

User Input: "Hey! Would you take \$50 for this old radio? It's pretty scratched up."

Agent Response: "Scratched? That's character, kid! You tourists don't know quality... [$\mathcal{M}_{char}$: Style Constraint + Annoyed Mood]. Wait, aren't you the one who **adopted that Art Deco lamp** last month? [Retrieve $\mathcal{M}_{user}$: Purchase History]. Okay, look. Since I know you're not just gonna flip it on eBay [Check $\mathcal{M}_{user}$: Trust Level], and honestly, I'm tired of dusting it [Reflect $\mathcal{M}_{char}$: Item State], I'll give it to you for \$60. But you treat it right!"

Table 4: Dual-Stream Data Construction Schema. This layout illustrates end-to-end data samples required for advancing memory-aware RPAs. Each case explicitly bifurcates the context into Character-Side Memory ($\mathcal{M}_{char}$) and Interaction-Side Memory ($\mathcal{M}_{user}$). Color-coded annotations in the dialogue explicitly demonstrate how cross-stream reasoning—fusing static character constraints with dynamic user history—causally drives the agent's behavior and conflict resolution.